Problem Set: Inference

Monday, October 19, 2026 9:36 PM

Use two-sided large-sample Normal tests and intervals throughout. Use the following critical values unless a problem says otherwise:

Confidence Level	Significance Level α	Two-Sided Critical Value
90%	0.10	1.645
95%	0.05	1.96
99%	0.01	2.576

For Problems 4-6, assume independent random samples from the two groups and independent observations within each group.

1. A random sample has

$$\bar{X} = 52, \qquad s = 10, \qquad n = 100.$$

Compute the estimated standard error of $\bar{X}$. Test

$$H_0 : \mu = 50 \qquad \text{vs.} \qquad H_1 : \mu \neq 50$$

at significance levels $\alpha = .10, .05, .01$ using the large-sample z-statistic. These correspond to 90%, 95%, and 99% confidence levels. Construct a 95% confidence interval for μ.

$SE(\bar{X}) = \frac{10}{\sqrt{100}} = \boxed{1}$

$CI = 52 \pm 1.96(1) = (50.04, 53.96)$

95% confident the mean falls between 50.04 to 53.96, ∴ we can reject H_0

2. Suppose

$$\hat{S} = 3.1, \qquad \widehat{SE}(\hat{S}) = 1.3.$$

Construct 90%, 95%, and 99% confidence intervals, and test the null hypothesis that $S = 0$ at the corresponding significance levels $\alpha = .10, .05, .01$. When do you reject or fail to reject the null of $S = 0$?

$\alpha = 0.10$: $CI = 3.1 \pm 1.645(1.3) = (0.962, 5.239)$

$\alpha = 0.05$: $CI = 3.1 \pm 1.96(1.3) = (0.552, 5.648)$

$\alpha = 0.01$: $CI = 3.1 \pm 2.576(1.3) = (-0.249, 6.449)$

Reject null of $S=0$ for $\alpha = 0.10, 0.05$, but fail to reject null for $\alpha = 0.01$, as it contains 0 in the interval.

3. A health survey compares a biomarker to a benchmark value of 12. The sample has

$$\bar{X} = 11.5, \qquad s = 3, \qquad n = 36.$$

Is 12 a plausible value for the population mean? Construct a 95% confidence interval for $H_0 : \mu = 12$ versus $H_1 : \mu \neq 12$. Do you reject or fail to reject the null?

$SE(\bar{X}) = \frac{3}{\sqrt{36}} = 0.5$ (6)

$CI = 11.5 \pm 1.96(0.5) = (10.52, 12.48)$

12 is a plausible value as it lies within the CI which says we are 95% confident the mean lies between 10.52 to 12.48, ∴ we fail to reject the null.

4. A real estate analyst compares the average price of a house in two neighborhoods. Group A:

$$\bar{X}_A = 25, \qquad s_A = 8, \qquad n_A = 100$$

Group B:

$$\bar{X}_B = 22, \qquad s_B = 7, \qquad n_B = 100.$$

Compute the two-sample Z-statistic for testing equality of means. Construct an approximate 95% confidence interval for $\mu_A - \mu_B$.

$SE(\mu_A - \mu_B) = \sqrt{\frac{8^2}{100} + \frac{7^2}{100}} = 1.063$

$Z = \frac{25 - 22}{1.063} = \boxed{2.822}$

$CI = (25 - 22) \pm 1.96(1.063) = \boxed{(0.916, 5.084)}$

We are 95% confident the mean difference between neighborhoods A & B is between 0.916 & 5.084.

5. A hospital compares recovery time under two treatments. Treatment A:

$$\bar{X}_A = 14, \qquad s_A = 6, \qquad n_A = 81$$

Treatment B:

$$\bar{X}_B = 12, \qquad s_B = 5, \qquad n_B = 64.$$

Compute the standard error of the difference in means and the corresponding test statistic for

$$H_0 : \mu_A - \mu_B = 0.$$

$$SE(\bar{X}_A - \bar{X}_B) = \sqrt{\frac{6^2}{81} + \frac{s^2}{64}} = \boxed{0.914}$$

$$z = \frac{14-12}{0.914} = \boxed{2.189}$$

6. A product-quality team compares defect scores across two factories. Factory A:

$$\bar{X}_A = 3.2, \qquad s_A = 1.5, \qquad n_A = 49$$

Factory B:

$$\bar{X}_B = 2.8, \qquad s_B = 1.4, \qquad n_B = 64.$$

Is the difference statistically large relative to the noise? Compute the test statistic and confidence interval for $H_0 : \mu_A - \mu_B = 0$, and interpret the result at $\alpha = .05$.

$$SE(\bar{X}_A - \bar{X}_B) = \sqrt{\frac{(1.5)^2}{49} + \frac{(1.4)^2}{64}} = 0.277$$

$$z = \frac{3.2-2.8}{0.277} = \boxed{1.446}$$

$$CI = (3.2-2.8) \pm 1.96(0.277) = \boxed{(-0.143, 0.943)}$$

We are 95% confident the difference in means of factories A & B is between (-0.143, 0.943).
B/c 0 is in the CI, we fail to reject the null $H_0, \mu_A - \mu_B = 0$.
The difference is not statistically large relative to the noise due to 0 in the 95% CI.

```python
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
import seaborn as sns
from scipy import stats
```

✓ 0.0s Python

7. In `./data/ames_prices.csv`, create an indicator that equals 1 when `Central.Air == 'Y'` and 0 otherwise.

a. Estimate the proportion of homes with central air.

```python
df = pd.read_csv('./data/ames_prices.csv')
df['Central.Air'].value_counts(dropna=False)
```

✓ 0.0s Python

```
Central.Air
Y    2734
N     196
Name: count, dtype: int64
```

```python
x = (df['Central.Air'] == 'Y').astype(int).to_numpy()
n = len(x)
p_hat = x.mean()

print(p_hat)
```

✓ 0.0s Python

```
0.9331058020477816
```

93.31% of homes have central air

b. Using a reproducible random-number seed, generate at least 2,000 bootstrap estimates of this proportion. Report the seed and number of bootstrap replications.

```python
seed = 42
B = 2000
rng = np.random.default_rng(seed)

boot = np.array([rng.choice(x, size=n, replace=True).mean() for _ in range(B)])
print(len(boot))
```

✓ 0.1s Python

```
2000
```

The seed is 42 and the number of boostrap replications is 2000

c. Compute the bootstrap standard error as the sample standard deviation of the bootstrap estimates (using `ddof=1` ensures that there's the $1/(N-1)$ degrees of freedom correction).

```python
se_boot = boot.std(ddof=1)
print(se_boot)
```

✓ 0.0s Python

```
0.004590333440769273
```

d. Using the original estimate and the bootstrap standard error, construct Normal 90%, 95%, and 99% confidence intervals.

```python
for conf in [0.90, 0.95, 0.99]:
    z = stats.norm.ppf(1 - (1 - conf) / 2)
    lo, hi = p_hat - z * se_boot, p_hat + z * se_boot
    print(f"{conf}: z={z:.3f}  [{lo:.4f}, {hi:.4f}]")
```

✓ 0.0s Python

```
0.9: z=1.645  [0.9256, 0.9407]
0.95: z=1.960  [0.9241, 0.9421]
0.99: z=2.576  [0.9213, 0.9449]
```

e. Use each confidence interval to evaluate $H_0 : p = .90$. At the matching significance levels $\alpha = .10, .05, .01$, state whether you reject or fail to reject the null hypothesis. Briefly interpret your results.

e. Use each confidence interval to evaluate $H_0 : p = .90$. At the matching significance levels $\alpha = .10, .05, .01$, state whether you reject or fail to reject the null hypothesis. Briefly interpret your results.

p = .90 is not within any of the confidence intervals at any of the significance levels, so we reject the null in every case, meaning there is strong evidence that the percent of homes with central air is above 90%.

8. In `./data/ames_prices.csv`, estimate the Pearson correlation between `np.arcsinh(area)` and `np.arcsinh(price)`.

   a. Using a reproducible random-number seed, generate at least 2,000 bootstrap estimates of the correlation by resampling paired rows. Report the seed and number of bootstrap replications.

```python
df = pd.read_csv('./data/ames_prices.csv')
df[['area', 'price']].isna().sum()

a = np.arcsinh(df['area'].to_numpy())
y = np.arcsinh(df['price'].to_numpy())
n = len(a)

r_hat = np.corrcoef(a, y)[0, 1]

print(n)
print(r_hat)
```

✓ 0.0s Python

```
2930
0.7231661530091357
```

```python
seed = 43
B = 2000
rng = np.random.default_rng(seed)

boot = np.empty(B)
for b in range(B):
    idx = rng.integers(0, n, size=n)
    boot[b] = np.corrcoef(a[idx], y[idx])[0, 1]
print(len(boot))
```

✓ 0.4s Python

```
2000
```

The seed is 43 and the number of bootstrap replications is 2000

b. Compute the bootstrap standard error as the sample standard deviation of the bootstrap estimates (using `ddof=1` ensures that there's the $1/(N-1)$ degrees of freedom correction).

```python
se_boot = boot.std(ddof=1)
print(se_boot)
```

✓ 0.0s Python

```
0.01013387836286545
```

c. Using the original estimate and the bootstrap standard error, construct centered-Normal 90%, 95%, and 99% confidence intervals.

```python
for conf in [0.90, 0.95, 0.99]:
    z = stats.norm.ppf(1 - (1 - conf) / 2)
    lo, hi = r_hat - z * se_boot, r_hat + z * se_boot
    print(f"{conf}: z={z:.3f}  [{lo:.3f}, {hi:.3f}]")
```

✓ 0.0s Python

```
0.9: z=1.645  [0.706, 0.740]
0.95: z=1.960  [0.703, 0.743]
0.99: z=2.576  [0.697, 0.749]
```

d. Use each confidence interval to evaluate $H_0 : \rho = 0$. At the matching significance levels $\alpha = .10, .05, .01$, state whether you reject or fail to reject the null hypothesis. Briefly interpret the association without making a causal claim.

p = 0 is not within any of the confidence intervals at any of the significance levels, so you would reject the null in every case. This suggests that larger homes in this sample tend to sell for more compared to the smaller homes.

e. Briefly explain why `area` and `price` must remain paired during resampling.

Area and price come from the same observation, so the bootstrapping has to perform a similar process to how the data was sampled for each observation, meaning we need to ensure they remain paired during resampling.

9. In `./data/metabric.csv`, use `Overall Survival (Months)` to estimate the mean recorded time to death or last follow-up. Include both living and deceased patients.

a. Using a reproducible random-number seed, generate at least 2,000 bootstrap estimates of the mean. Report the seed and number of bootstrap replications.

```python
df = pd.read_csv('./data/metabric.csv')
col = 'Overall Survival (Months)'

df[col].isna().sum()
df['Overall Survival Status'].value_counts()
```

✓ 0.0s Python

```
Overall Survival Status
1:DECEASED    768
0:LIVING      575
Name: count, dtype: int64
```

```python
x = df[col].to_numpy()
n = len(x)
mu_hat = x.mean()

print(mu_hat)
```

✓ 0.0s Python

```
129.09429138744105
```

```python
seed = 44
B = 2000
rng = np.random.default_rng(seed)

boot = np.array([rng.choice(x, size=n, replace=True).mean() for _ in range(B)])
print(len(boot))
```

✓ 0.1s Python

```
2000
```

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The seed is 44 and the number of bootstrap replications is 2000

b. Compute the bootstrap standard error as the sample standard deviation of the bootstrap estimates (using `ddof=1` ensures that there's the $1/(N-1)$ degrees of freedom correction).

```python
se_boot = boot.std(ddof=1)
print(se_boot)
```

✓ 0.0s Python

```
2.1065533711642814
```

c. Using the original estimate and the bootstrap standard error, construct centered-Normal 90%, 95%, and 99% confidence intervals.

```python
for conf in [0.90, 0.95, 0.99]:
    z = stats.norm.ppf(1 - (1 - conf) / 2)
    lo, hi = mu_hat - z*se_boot, mu_hat + z*se_boot
    print(f"{conf}: [{lo:.2f}, {hi:.2f}] months")
    print(f"{conf}: [{lo/12:.2f}, {hi/12:.2f}] years")
```

✓ 0.0s Python

```
0.9: [125.63, 132.56] months
0.9: [10.47, 11.05] years
0.95: [124.97, 133.22] months
0.95: [10.41, 11.10] years
0.99: [123.67, 134.52] months
0.99: [10.31, 11.21] years
```

d. Use each confidence interval to evaluate $H_0 : \mu = 120$ months. At the matching significance levels $\alpha = .10, .05, .01$, state whether you reject or fail to reject the null hypothesis.

μ = 120 is not within any of the confidence intervals at any of the significance levels, so we can reject the null in each case.

e. Interpret the estimate and confidence intervals in both months and years. Because living patients' eventual death times are right-censored, explain why these results do not estimate unrestricted mean lifetime.

For confidence interval in years, we just divide each interval by 12 to find the years, so they would then be as follows: 0.9: [10.47, 11.05] years, 0.95: [10.41, 11.10] years, and 0.99: [10.31, 11.21] years. Still, we reject the null in each case, as none of them contain 10 years in their intervals. However, they do not estimate the unrestricted mean lifetime because the recorded times represent how long they have been observed, rather than when they will eventually die, so their actual lifetimes may be longer than the recorded values.

10. In `./data/nhanes_data_17_18.csv`, select respondents who are at least 18 years old and have an observed value for `SleepHoursWeekdaysOrWorkdays`. Estimate their mean weekday sleep.

a. Using a reproducible random-number seed, generate at least 2,000 bootstrap estimates of the mean. Report the seed and number of bootstrap replications.

```python
df = pd.read_csv('./data/nhanes_data_17_18.csv', low_memory=False)
col = 'SleepHoursWeekdaysOrWorkdays'
age = 'AgeInYearsAtScreening'

d = df[(df[age] >= 18) & df[col].notna()]
x = d[col].to_numpy()
n = len(x)
mu_hat = x.mean()
print(mu_hat)
```

4] ✓ 0.3s Python

```
7.62124248496994
```

```python
seed = 45
B = 2000
rng = np.random.default_rng(seed)

boot = np.array([rng.choice(x, size=n, replace=True).mean() for _ in range(B)])
print(len(boot))
```

5] ✓ 0.1s Python

```
2000
```

The seed is 45 and the number of bootstrap replications is 2000

b. Compute the bootstrap standard error as the sample standard deviation of the bootstrap estimates (`ddof=1`).

```python
se_boot = boot.std(ddof=1)
print(se_boot)
```

✓ 0.0s Python

```
0.023299423677025366
```

c. Using the original estimate and the bootstrap standard error, construct centered-Normal 90%, 95%, and 99% confidence intervals.

```python
for conf in [0.90, 0.95, 0.99]:
    z = stats.norm.ppf(1 - (1 - conf) / 2)
    print(f"{conf}: [{mu_hat - z*se_boot:.3f}, {mu_hat + z*se_boot:.3f}]")
```

✓ 0.0s Python

```
0.9: [7.583, 7.660]
0.95: [7.576, 7.667]
0.99: [7.561, 7.681]
```

d. Use each confidence interval to evaluate $H_0 : \mu = 8$ hours. At the matching significance levels $\alpha = .10, .05, .01$, state whether you reject or fail to reject the null hypothesis.

μ = 8 is not within any of the confidence intervals at any of the significance levels, so we would reject the null in every case.

e. Interpret the result and discuss whether its statistical and practical significance tell the same story. For this exercise, treat observations as independent and unweighted. Because the complex survey design is ignored, do not describe the result as nationally representative or the intervals as survey-adjusted.

Adults in this sample report an average of about 7.62 hours of sleep during a weekday. After doing calculating confidence intervals, we determined that we reject the null of 8 hours of sleep for each significance level, so the result is statistically significant due to such a low standard error. The practical signficance is different, however, because the confidence intervals are only about 25 minutes below 8 hours at the high end, so there is not much difference between the results we obtained and the actual 8 hours of sleep.